

One-Dimensional Interior Models of Europa

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1 Introduction

Europa is one of Jupiter’s largest moons. It is a scientifically interesting planetary body, as it is commonly thought to have a subsurface liquid water ocean that may be capable of harbouring life. In this report, various models of Europa’s interior found in literature are analysed and compared. Based on these, three new models are constructed, with the aim of characterising Europa’s global interior.

2 Literature Study

Several existing models of Europa’s interior were consulted for the purposes of this assignment. Jin and Ji[1] published an article in which they outline the creation of 3 models of Europa, the difference between the models being the assumed composition of the core: Fe, FeS, or $\text{Fe}_{0.8}(\text{FeS})_{0.2}$. Aside from the core, Jin and Ji define two more layers in each of their models: an olivine mantle and an outer shell made of water. To generate interior profiles of Europa, they choose a fractional mass for each of the 3 layers as well as a centre pressure. The program they developed performs an integration from the centre to the surface in order to solve for the surface pressure, logically must be 0. Based on the calculated surface pressure, the program then adjusts the centre pressure before restarting the integration, repeating this sequence until the calculated surface pressure is 0.

Another existing model found in literature is one published by Cammarano et al.[2]. The aim of their research was to find the seismic velocities associated with different possible interior of Europa, so their model determines the density profile based on temperature and composition. Like Jin and Ji, Cammarano et al. assume an FeS core and a water shell, but unlike Jin and Ji, they consider two different mantle compositions: pyrolitic and chondritic.

Kuskov and Kronrod[3] base their model on the observed mass, bulk density, and moment of inertia of Europa. Unlike the two models perviously discussed in this section, in this model the mantle is divided into 3 layers, leading to 5 layers in total.

Finally, Howell[4] focuses on Europa’s icy outer shell, and attempts to determine the thickness of this shell based

on its steady-state heat flux. This model differs from the others in that it does not provide a profile of Europa’s full interior, but rather provides a detailed profile of the ice crust based on assumed values for the rest of the interior. Because all models found use a different approach and were created with a different goal in mind, a direct comparison is difficult to make. Nevertheless, there are a few notable trends among the articles. There seems to be a general consensus among researchers that Europa consists of an iron-rich core, a silicate or chondritic mantle, and an ice/water outer shell¹. Numerically, there is some variation between the outputs of each model, part of which can be explained by the fact that the models use different theoretical values as inputs, most of which are, themselves, obtained through theoretical modeling². A summary of relevant values is given in Table 1. As each of these models has undergone some form of verification and validation before being published, these values, or averages thereof, will be used as reference values for the remainder of this report.

3 General Modeling Approach

For this report, three one-dimensional models of Europa’s interior were created. Models I and II are written from scratch in Python, with model II being an extension of the simpler model I. Model III is written using the BurnMan Python module, and the setup of this model will be described in Section 7.

An overview of the input and output parameters used in model I is given in Table 2. The approximate density of the core, mantle, and outer shell are taken from Jin and Ji[1]. The simulation starts by upward integrating the mass for each step in radius and calculating the gravitational acceleration at each of those radii, according to Equations 1 and 2, respectively. Then, the simulation integrates down-

¹It should be noted, though, that this is by no means definitively proven. See for example Trinh et al.[5], in which the existence of Europa’s metallic core is questioned based on reanalysis of Galileo data.

²A frequently cited paper in literature about Europa’s interior is Anderson et al. (1998)[6], which is where a lot of input values are taken from. Unfortunately, this article appears to be behind a paywall.

Table 1: An overview of the models analysed for this assignment.

	Jin and Ji	Cammarano et al.	Kuskov and Kronrod	Howell
Total radius [km]	1563	Not provided	1565	1561
Core radius [km]	777	~600	880	630
Shell thickness [km]	167	~100	105-160	127
Moment of inertia [-]	0.330	0.346	0.346	0.359
Total mass [kg]	$4.80 \cdot 10^{22}$	$4.8 \cdot 10^{22}$	$47.99 \cdot 10^{21}$	$4.80(10)^{22}$
Centre pressure [GPa]	~5.5	Not provided	~4.8	Not provided

ward according to Equation 3 to find the pressure at each radius step.

$$\frac{dM}{dr} = 4\pi\rho r^2 \quad (1)$$

$$g(r) = \frac{GM(r)}{r^2} \quad (2)$$

$$\frac{dp}{dr} = -\rho g(r) \quad (3)$$

Once this is done, the dimensionless moment of inertia is calculated, using Equation 4.

$$\bar{I} = \frac{8\pi}{3MR^2} \int_0^R \rho(r)r^4 dr \quad (4)$$

This is then compared to the observed dimensionless moment of inertia. If the calculated moment of inertia is higher than the observed value, the shell thickness is increased, and if it is lower, the shell thickness is decreased. The total calculated mass is also compared with the observed mass, and if the calculated mass is higher than the observed value, the core radius is decreased, and vice versa. The simulation is run again with the adjusted values, and it continues in this loop until both the calculated mass and moment of inertia are within 1% of the observed value. Model II works in the same way, except the density values are now dependent on temperature and pressure.

Table 2: An overview of the input and output parameters for model I.

Inputs	Outputs
Observed radius	Mass
Observed mass	Gravity
Core density	Pressure
Mantle density	Moment of inertia
Shell density	Core radius
Step size	Shell thickness
Initial core radius	
Initial shell thickness	

4 Results, Verification and Validation of Model I

To construct model I, the following assumptions were made:

- Europa consists of three layers: an FeS core, a pyroclitic mantle, and an outer shell of water and ice;
- Because it is expected to be relatively thin compared to the rest of the water shell, the ice crust is neglected and the outer shell is assumed to be 100% liquid water[1];
- The density of each layer is homogeneous and is 5500, 3300, and 1000 kg/m³ for the core, mantle, and outer shell, respectively. These values were estimated from the graphs published by Jin and Ji[1];
- Europa's observed radius is 1565 km;
- Europa's observed mass is $479.7(10)^{20}$ kg;
- Europa's observed dimensionless moment of inertia is 0.346.

The initial core radius and outer shell thickness are chosen somewhat arbitrarily: based on the values found in Table 1, the core-mantle boundary is initially set to 800 km from the centre, and the mantle-shell boundary is set to 1400 km from the centre, which gives an outer shell thickness of 165 km. Running the program with these inputs, using a step size of 1 km, results in a predicted mass of $4.84(10)^{22}$ kg, which deviates 0.96% from the observed value, and a moment of inertia of 0.346, which deviates 0.031% from the observed value. Furthermore, the model predicts the pressure at the centre of Europa to be 5.1 GPa, which roughly agrees with the values in Table 1. The gravitational acceleration at the surface is found to be 1.32 m/s², which matches the value 1.31 m/s² given by Cammarano et al.[2]. The core radius and shell thickness found are 683 and 116 km, respectively, most closely resembling the models by Cammarano et al.[2] and Howell[4]. It takes the program 137 iterations to converge to these values.

To verify the model, a few tests were done. The sensitivity of the code was checked by decreasing the initial core

radius by 1 km and running the program again. Another check was done by decreasing the initial shell thickness, and two more by increasing each of the values by 1 km. All output values remained the same in each of these four cases. Even when the initial core radius is set to 0, the program eventually converges to roughly the same values as with more realistic initial conditions. Together with the fact that the functions written for the simulation were verified by comparing their outputs with analytical solutions of the equations they were made to solve, this model can be considered verified.

The model was validated by changing all the inputs to Earth’s characteristics, as these are known with more certainty than Europa’s characteristics³. The model accurately predicted the pressure at Earth’s centre as well as the gravitational acceleration at Earth’s surface. It also quite accurately predicted the size of Earth’s core, but, interestingly enough, it seems to favour an unrealistically thick crust regardless of the initial conditions set. This could be a consequence of the somewhat oversimplified criteria for adjusting the core radius and crust thickness, or of the assumption that the density of each layer is homogeneous. Such an assumption should more significantly affect a model of Earth than of Europa simply because Earth is so much more massive. It can be concluded that although it is a simplified model, it is accurate and realistic enough for the purpose of this report.

5 Thermal Effects

To construct model II, model I is extended to include the effects of temperature on the density of each of Europa’s layers. To this end, three thermal profiles are constructed, based on three centre temperature values found in literature: a minimum temperature, a maximum temperature, and a mean temperature.

Information about the core temperature of Europa is not easy to find. When it is mentioned in literature, it appears to be mostly speculative values, which makes sense, given the limited information available about Europa in general. Cammarano et al.[2], for example, argue that for the core to be solid, its temperature must be below 1900 K for a pure iron core, and below 1400 for an FeS core. Travis et al.[7] propose a core temperature of 1750 K based on their simulations of the thermal history of Europa. Kuskov and Kronrod[8] found a temperature of 1293 degrees Celsius, or 1566 K, at the core-mantle boundary using one of their simulations. Finally, Prentice[9] modeled Europa with a core temperature of 1073 K.

Based on these values, three core temperatures were selected, as shown in Table 3. Using a linear temperature profile ranging from the core temperature to the surface temperature, chosen to be 104 K[4], a density profile can

Table 3: The centre temperature values chosen as basis of the three thermal profiles for Europa.

Min	1073 K
Mean	1447 K
Max	1750 K

Table 4: The values used to determine the density at each point in the profile.

	Core	Mantle	Shell
Material	Iron sulfide	Pyrolitic	Water
Density [kg/m ³]	5500	3300	1000
Expansivity [K ⁻¹]	87e-6[10]	40e-6[11]	-4.93e-8[12]
Bulk modulus [GPa]	75.2[13]	130[14]	10.56[15]

be constructed using the following equation:

$$\rho(T) = \rho_0(1 - \alpha_T \Delta T + \frac{1}{K} \Delta p) \quad (5)$$

In the above equation, α_T represents the thermal expansivity of the material, and K represents the bulk modulus of the material. ρ_0 represents a reference density. The values used in model II to calculate the density as a function of the temperature are provided in Table 4. Equation 5 is implemented into model I by using it to update the density profile after each iteration. Because Equation 5 requires the pressure as an input, and the density profile is required to calculate the pressure profile, the first iteration is done using the same homogeneous density profile used in Model I. Aside from these things, the model works in the same way as model I.

6 Results, Verification and Validation of Model II

The results of model II are summarised in Table 5. To validate model II, the resulting graphs were qualitatively compared with graphs found in literature. Comparing model II to the model created by Jin and Ji[1] shows that the shape of the plot is very similar to that plotted by Jin and Ji. It is interesting to note that although the shapes of the plots are quite similar, the centre pressure obtained by model II is lower than any of Jin and Ji’s models by almost 0.5 GPa. Comparing model II to the values given in Table 1, all values roughly match those found in literature.

7 Results and Discussion of Model III

Model III makes use of the Burnman module available for Python and was created by modifying the file “tutorial.03.layers_and_planets.ipynb” included in the Burnman installation. The Burnman module includes a minimal library, but it is somewhat limited. Thus, for model

³The thickness and density of each layer was taken from <http://hyperphysics.phy-astr.gsu.edu/hbase/Geophys/earthstruct.html>.

Table 5: The results of model II for the minimum core temperature, mean core temperature, and maximum core temperature.

	Min	Mean	Max
Mass deviation	0.976%	0.941%	0.956%
Centre pressure [GPa]	4.923	4.921	4.921
Gravity at surface [m/s^2]	1.294	1.295	1.295
Core radius [km]	654	666	677
Shell thickness [km]	125	123	120
MOI deviation	0.0356%	0.0350%	0.0256%
Iterations	54	66	77

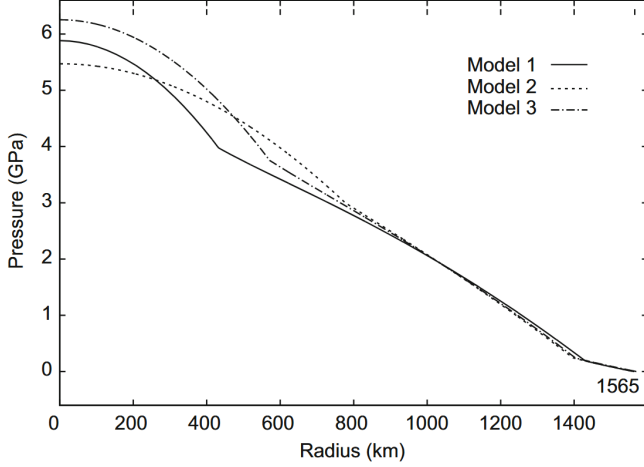


Figure 1: The interior pressure of Europa as determined by Jin and Ji[1].

III, slightly different materials were chosen than for models I and II: the core is assumed to be pure iron rather than iron(II)sulfide, the mantle is assumed to consist of Olivine, and the outer shell is assumed to be entirely liquid water. For each layer, the same linear temperature variation is used as in model II, and the pressure mode is set to “self-sufficient”. The initial values for pressure and gravity are taken from the results of model II (at the mean core temperature), and so are the layer thicknesses. The results of this model, as well as a comparison to models I and II, are shown in Figure 2. An overview of some key parameters computed by model III are shown in Table 6. This table shows that while model III manages to get close to the observed moment of inertia factor regardless of the temperature profile, the deviation in mass is rather large.

Table 6: The results of model III for the minimum core temperature, mean core temperature, and maximum core temperature.

	Min	Mean	Max
Mass deviation	16.23%	16.58%	17.07%
MOI deviation	0.0286%	0.0934%	0.0282%
Iterations	5	5	5

8 Comparison Between Models

When viewing Figure 2, one difference between the models immediately stands out. While models I and II match almost perfectly, model III consistently computes higher values for density, pressure, and gravity. One way this could be explained is the difference in assumed layer compositions. In the core, the figure shows the density found in model III is over 25% higher than that for models I and II. This makes sense when considering that the core is assumed to consist of iron (II) sulfide (FeS) in models I and II and of pure iron (Fe) in model III. By comparing the atomic weights of iron and sulfur it logically follows that pure iron would have a higher density. Since gravity and pressure are both functions of density, this difference, along with the smaller but still significant differences in mantle and shell density, can account for the differences in gravity and pressure found.

As mentioned before in Section 2, some models found in literature do assume a core made of pure iron, so this is an avenue worth exploring. As seen in Section 7, the moment of inertia factors found by model III closely match the observed value, but the mass found is significantly larger than the observed one. This implies one of two things: Europa’s core is less dense than one made of pure iron, or Europa’s core is smaller than it is assumed to be in model III. Running an iterative version of model III (in much the same way as was done for model II) leads to the program decreasing the core radius to 0 without actually converging to the observed mass, while the moment of inertia factor converges to the observed value immediately. One could conclude from this that Europa’s core cannot be made of pure iron as it is too dense to fit both the observed mass and moment of inertia factor. However, it should be noted that there are some flaws in this iterative model. Most glaringly, the pressure and gravity values at the layer boundaries are not iterated. Further research is needed to find out if this conclusion is actually valid or not.

9 Conclusion

In this report, four interior models of Europa from literature were analysed and compared to each other. Furthermore, three new models were constructed and compared to both each other and the models found in literature. The first two models have shown to provide a decent first estimate for the interior profile of Europa, and they roughly agree with those found in literature. However, there is a lot of uncertainty in the input values taken from literature. Future missions to Europa could provide more accurate data that can be used to construct more accurate models. The third model agrees with the observed moment of inertia factor, but not with the observed mass. This model could in the future be modified to contain a more accurate layer distribution. The models are quite simplified, and neglect many factors necessary to create

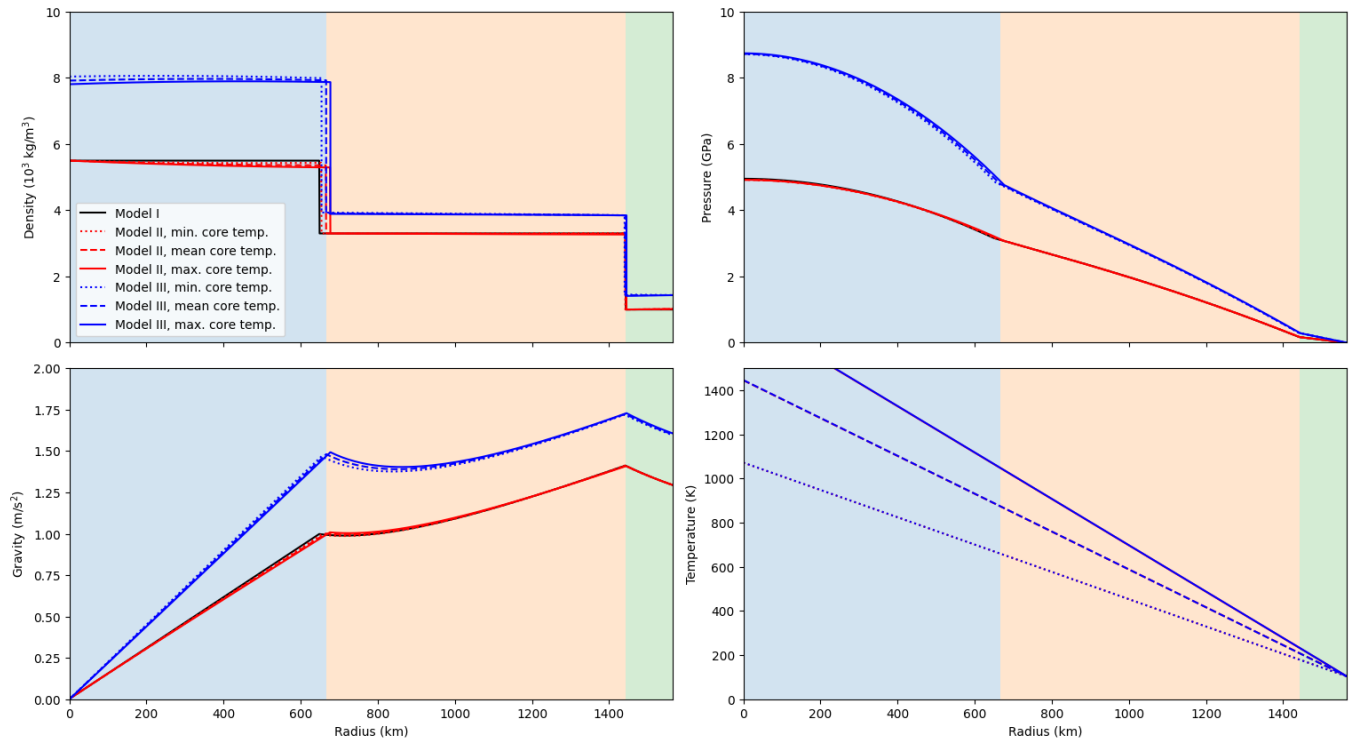


Figure 2: The density, pressure, gravity, and temperature profiles of Europa as calculated by the 3 different models.

a realistic model. In future research, these models could perhaps be expanded to include more than 3 layers, and thermal effects such as tidal heating or radiogenic heating.

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